

A Descriptive Comparison of Oral Cavity Movements Between Brass Instrumentalists Performing Large Interval Slurs

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OBJECTIVE: We have previously described patterns of oral cavity movement in advanced French horn players during the performance of large interval slurs. The current study expands upon that work by reporting comparisons between horn, trumpet, and trombone players performing similar large interval exercises. **METHODS:** Real-time MRI (RT-MRI) films at 20 msec resolution were simultaneously obtained in the sagittal and coronal planes in 10 trumpet players, 9 horn players, and 10 trombone players as they performed 2 repetitions each of slur sequences spanning 1 octave and 1 octave + 3rd (interval of a tenth) at a *mezzo forte* dynamic level. Nine profile lines were overlaid on the sagittal images allowing the measurement of dorsal tongue edge movement using a customized MATLAB toolkit. Image J (FIJI) was utilized to measure the cross-sectional area formed between the dorsal edge of the tongue and the hard palate as depicted in coronal images. **RESULTS:** In horn and trumpet players, but not in trombone players, sagittal and coronal views demonstrate patterned tongue movements that narrow and widen the air channel during ascending and descending slurs, respectively. The magnitude of these movements is greater during larger intervals, though not perfectly consistent within each slur sequence. The sagittal views revealed these patterns to be most prominent in the anterior oral cavity rather than the middle and posterior oral cavity across all instruments. **CONCLUSION:** Conformational changes of the oral cavity (vocal-tract tuning) during large interval slurs in the upper register may assist performance in trumpet and horn players. Trombone players may also employ these strategies when the extreme upper register is involved, but future study must verify this. *Med Probl Perform Art* 2024;39(4):169-176.

KEYWORDS: musician, brass instrument, tongue movement, real-time MRI

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UNDERSTANDING the conformational changes within the oral cavity associated with brass performance has long been a goal of brass pedagogues over the years. Some of the earliest studies attempting to describe these movements were conducted in the 1960s and 1970s involving the use of fluoroscopic videography in trumpet players.^(1,2) While these projects provided pioneering and seminal information for brass teachers and students, they present limitations to investigators desiring to expand such work. First, the image quality obtained is of rather poor spatial resolution, and second, fluoroscopic techniques involve the use of radiation which can be dangerous for the performer, thus interfering with systematic widespread use in the study of musicians.

A practical solution that was introduced in 2010 involves the use of real-time magnetic resonance imaging (RT-MRI) technology.⁽³⁾ This was first applied to the study of brass playing by Schumacher et al.⁽⁴⁾ in trumpet performers, and since that time has become more widely applied. The first paper published by our research group in 2015 using RT-MRI was a methodological pilot study comparing lingual motor function within the oral cavities of a tuba, trombone, horn, and trumpet player during the performance of an ascending harmonic sequence of notes, and a double-tonguing trial.⁽⁵⁾ The exercises were performed on a plastic practice device known as a buzz extension and resistance piece (B.E.R.P., Musical Enterprises, Fairfax, CA) using plastic mouthpieces. This study was conducted primarily as a “proof-of-concept” paper and affirmed the utility of using both high-speed RT-MRI video (30 Hz acquisition rate) and a customized MATLAB toolkit to describe and quantify conformational changes within the oral and pharyngeal cavities during brass performance.

Since that time, we have published numerous studies examining lingual movements in both healthy and embouchure dystonia subjects in which MRI-compatible instruments (horn, trumpet, trombone) are employed.⁽⁶⁻¹⁴⁾ These instruments were originally comprised of a plastic mouthpiece, plastic tubing, and an instrument-specific brass bell flare. The length of plastic tubing was precisely determined to produce a horn in E $\flat$, a trumpet in C, and a trombone in B $\flat$. Most recently, we have developed an MRI-compatible horn in F and E $\flat$, a trumpet in B $\flat$, and two B $\flat$ trombones (tenor and bass) comprised entirely of non-



FIGURE 1. Large interval slur exercise consisting of 2 intervals, each with 2 upward and 2 downward slurs. Pitches are depicted as they appear for horn in Eb. Actual concert pitches are Eb₃-Eb₄ (first interval series), and Eb₃-G₄ (second interval series).

ferromagnetic brass. This greatly enhances the acoustic properties and playing characteristics, making them very similar to regular instruments in performing exercises utilizing the natural harmonics that occur over multiple octaves.

The corpus of our work has fostered considerable interest in the field of brass pedagogy. Numerous presentations have been made at international conferences (International Horn Symposiums in 2015, 2016, 2018, 2019, International Trombone Festival 2019). Most noteworthy is the explicit use of images and films in the publication of a popular brass pedagogy book, *Horn Playing from the Inside Out: A Method For All Brass Musicians*, by Eli Epstein.⁽¹⁵⁾ The latest e-version of this book (Apple) uses many of our videos and images as an aide for teachers and students.

To date, we have published one study using the all-brass horn that examined lingual movements of the tongue during large interval slurs in advanced players.⁽¹⁴⁾ The results indicated that patterned tongue movements are employed which respectively narrow and widen the air channel during ascending and descending slurs.⁽¹⁴⁾ With the exception of our very first paper,⁽⁷⁾ however, no attempt has been made to make comparisons between the different brass instruments when performing similar exercises. The purpose of the current investigation, therefore, is to make inter-instrument comparisons of lingual movements during large interval slurs (octaves and tenths) between advanced trumpet, horn, and trombone performers utilizing all-brass instruments.

METHODS

Subjects and Performance Device

Ten trumpet, 9 horn, and 10 trombone players volunteered to participate in this study. All were professional-level performers and teachers recruited from orchestras and conservatories. The trumpet players were recruited from Denmark (n=1), Germany (8), and Belgium (1). Two of the horn players were from the United States, and the remaining 7 were from Germany. Trombone players came from the UK (2), and the remaining 8 were from Germany.

All testing was conducted at the Biomedical NMR facility at the Max-Planck-Institute for Multidisciplinary Sciences in Göttingen, Germany. Prior to testing, the subjects gave written informed consent in accordance with the recommendations of the local ethics committee in Göttingen.

The subjects performed on all-brass MRI-compatible instruments, described above. The horn and trumpet were designed and built by Richard Seraphinoff (Bloomington, IN, USA) while the trombone bell flares were designed and built by Yamaha Corporation (Hamamatsu, Japan), with the remaining brass tubing sections built by Richard Seraphinoff. The instruments lacked valves (horn and trumpet) or slides (trombone) and required all note changes to be accomplished by altering embouchure tension, flow rate, and oral cavity conformation. All musical exercises were performed in the supine position within the MRI magnet as described in our previous studies.^(6-8,12-14)

Testing Protocol

The data for this study were obtained over the course of several years (2015–2024) from different study protocols that have been employed in building the MRI Brass Repository.⁽¹⁶⁾ Across these protocols examining trumpet, horn and trombone players, octave and tenth intervals are commonly studied, though in the case of trombone players, intervals as large as two octaves are performed. From the repository, we therefore selected exercises representing these intervals for each instrument. This involved slurring up and down two times between Bb₃ and Bb₄, and Bb₃ and D₅ (trumpet); Eb₃ and Eb₄, and Eb₃ and G₄ (horn); and between Bb₂ and Bb₃, and Bb₂ and D₄ (trombone). Figure 1 illustrates an example as performed on horn (key of concert Eb).

Copies of the exercises were provided to the subjects prior to their scheduled test date, and 10–15 minutes of practice on the appropriate MRI-compatible instrument were provided at the laboratory on the day of each experiment. This included trials in the supine testing position inside the scanner prior to data acquisition. For the recorded trials, the subjects assumed the supine position inside the MRI magnet. The bell portion of each instrument was positioned outside the magnet at the feet of the subject, with the remaining brass tubing extended headward up into the scanner. The musicians held the mouthpiece and lead pipe during each trial. A fiber-optic microphone (Dual Channel-FOMRI; Optoacoustics; Or Yehuda, Israel) was positioned near the bell to record sound.

The details of how recordings were obtained have been previously explained⁽¹⁴⁾ and will only be summarized here. A PowerPoint presentation, operated by a technician from the scanner control room, regulated each trial. This

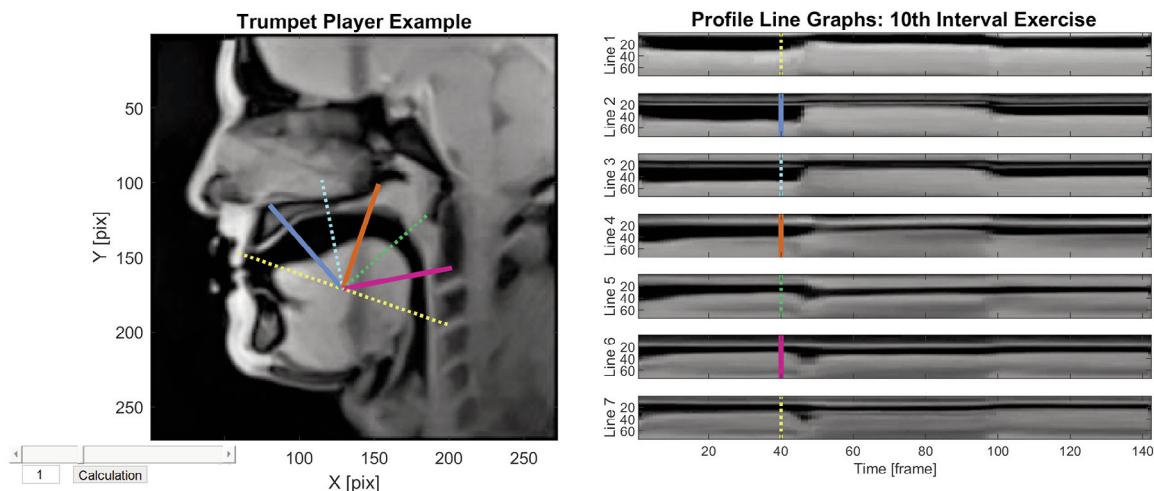


FIGURE 2. Profile line placement within the oral cavity (*left panel*) and resulting profile line graphs (*right panel*) in a representative subject. Lines 2, 4, and 6 representing the anterior, middle, and posterior oral cavity were selected for analysis. Vertical lines (*right panel*) are positioned prior to performing the top note.

presentation allowed musicians to visually see the exercise on a monitor in the scanner room, sequentially cued the performers throughout the trial, triggered the scanner and audio recorder to run for the prescribed time, and regulated performance tempo by interfacing with a flashing LED display.

An intercom system between the scanner and control rooms allowed communication between the subjects and the test administrator between each recording interval. During actual testing, if a trial was unsatisfactory to the subject or administrator, repeated trials were done, and the best performance trial was chosen. Within the selected 2-slur trial, the best slur iteration consisting of 1 upward and 1 downward slur was used for analysis.

Detailed descriptions of the methods and validation of RT-MRI have been previously published^(3,17,18) as well as an explanation of the method for obtaining dual-slice real-time MRI images.^(13,19) Briefly, the experiments were performed on a 3 T MRI system (Magnetom Prisma; Siemens Healthineers; Erlangen, Germany) with a maximum gradient strength of 80 mT/m. Data acquisition was accomplished with use of a 64-channel head coil. MRI measurements yielded an in-plane resolution of 1.4 mm, slice thicknesses of 8 mm, and a field-of-view of $192 \times 192 \text{ mm}^2$. Alternating sagittal and coronal image acquisitions at $2 \times 25 \text{ fps}$ (20 ms/frame) were interleaved to produce the dual-slice images and were subsequently sequenced and synchronized to audio recordings of each performance to produce the dual-slice films.

Data Analysis

The dual-slice films (simultaneous sagittal and coronal views) were separated for analyses, each film having identical audio tracks associated with it. Two quantitative analysis procedures were employed. The first of these methods has been used to study movements in the sagittal

view in numerous previous studies conducted by our group.^(5,6,8,11–14) To summarize, we make use of a custom MRI toolbox developed for MATLAB (MATLAB 2018a; MathWorks; Natick, MA, USA), as well as standard MATLAB image processing and signal processing toolboxes. Using these tools, a standardized array of 7 profile lines is produced, separated by 30° , originating roughly in the center of the tongue and radiating outward. Each profile line spans across the cavity created between the dorsal tongue surface and the lips, top of the oral cavity, or back of the throat (Figure 2a). Dynamic movements of the tongue along these profile lines are depicted in corresponding profile line graphs for each frame of the exercise (Figure 2b). For quantitative analyses of these movements, three of the profile lines were examined; lines 2, 4 and 6 corresponding to the anterior, middle and posterior oral cavities, respectively. Using the MATLAB custom MRI toolbox, specific time periods corresponding to the duration of each note are identified, and the average tongue edge position (pixels) for each of these notes is calculated and used to determine movement of the tongue between each note (see Results section).

The coronal films were analyzed using Image J, FIJI platform.⁽²⁰⁾ As in the case of the sagittal analyses, the best iteration of either the octave or tenth slur was studied for every performer. Three events for each slur were identified: the onset of the low note (the root), the onset of the highest note (the octave or tenth), and the onset of the repeated low note (the root). At each of these points, the outline of an air channel formed between the dorsal surface of the tongue and the roof of the mouth was clearly visible (Figure 3). For each iteration of the root note in the 3-note sequence, scrolling ahead in time from the moment of note onset revealed that a “settling” period was required before the image stabilized during the sustained note. The frame corresponding to the first stabilized position was used for measurement. With respect to the high note in each sequence, scrolling forward

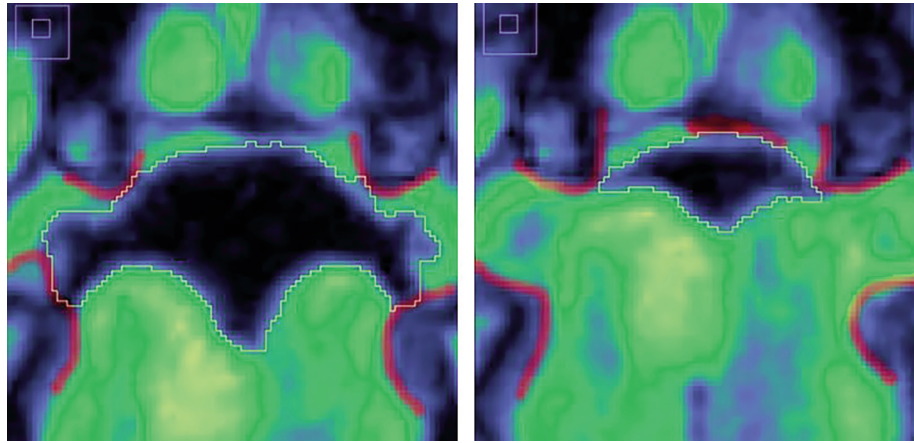


FIGURE 3. Outline of the supra-lingual air channel during performance of the root note (Eb_3 , left panel) and the octave above (Eb_4 , right panel) in a representative horn player (600X magnification). Red outlines are the upper and lower molars.

a few frames from the audible onset of this note revealed a moment when the air channel was at its smallest, and this frame was selected for measurement.

To aid in visualizing the edge between the air channel and tissue, a pseudo-color image look-up table in Image J (LUT, Green Fire Blue) was employed. This LUT application allows differences in luminescence intensity to be seen as differences in color rather than standard gray scale contrast (Figure 3). In our former study with horn players,⁽¹⁴⁾ we simply measured the vertical height of this channel as an indicator of area changes. However, in the current study, the actual area of the supra-lingual air channel (SLAC) was identified using the Image J wand tool (Legacy mode) with the maximum luminescence tolerance set to 60 for all images. This standardized the inclusion and exclusion of pixels within the SLAC area across all subjects and trials. When the upper and/or lower molars contacted the oral cavity, their borders were selected using a paint tool (see Figure 3, red outlines). This prevented the wand tool from

selecting dark pixels within the teeth as part of the air channel. The area of the air channel in pixels² was subsequently calculated using the Image J Analyze/Measure function.

RESULTS

Sagittal Image Analyses

Figure 4 depicts movement of the dorsal tongue surface along profile lines 2, 4 and 6 (anterior, middle and posterior oral cavities, respectively) during the ascending and descending phases of the octave slur. Columns above the baseline represent upward movement of the tongue edge and columns below the baseline indicate downward movements of the tongue edge. The greatest amount of movement was seen in the anterior oral cavity in the trumpet and horn players. The magnitude of upward movement for trumpet players is comparable to the downward movement (3.52 and -3.73 mm during the ascending and

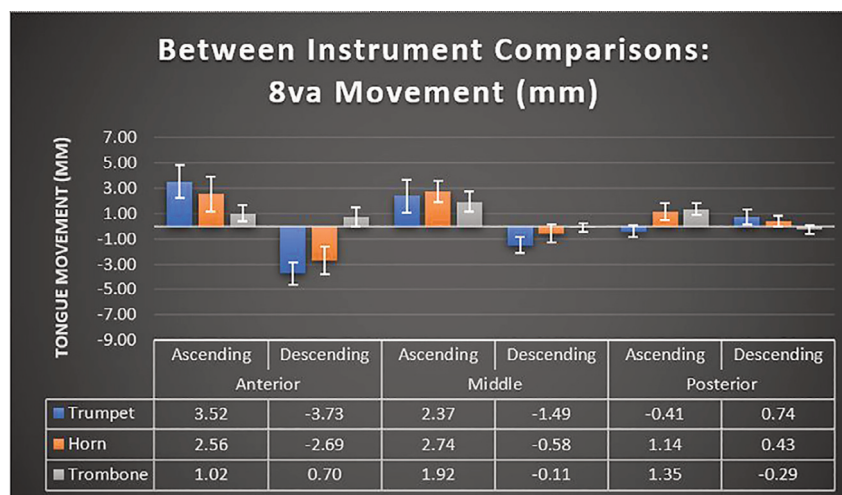


FIGURE 4. Dorsal tongue edge movement (mean $\pm$ SE) for each instrument in three regions of the oral cavity during ascending and descending phases of the octave slur exercise.

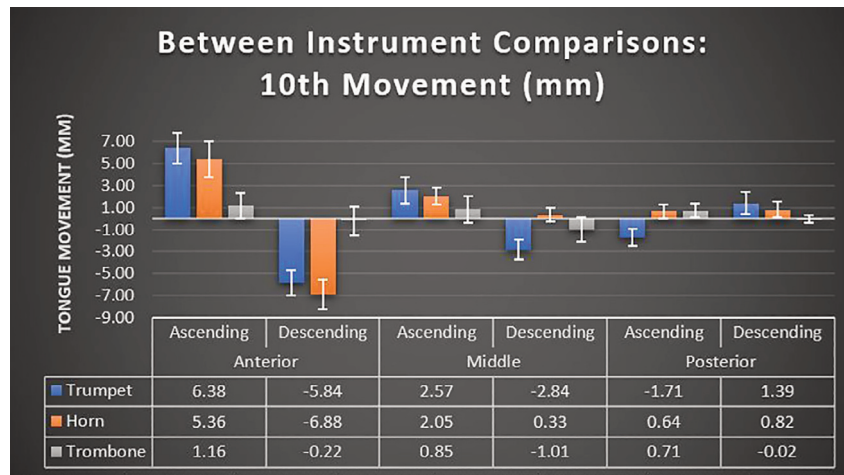


FIGURE 5. Dorsal tongue edge movement (mean $\pm$ SE) for each instrument group in three regions of the oral cavity during ascending and descending phases of the tenth interval slur exercise.

descending phases of the slur, respectively). A similar ascending and descending pattern is shown for the horn players, though the magnitude of movement is slightly less (+2.56 and -2.69 mm, respectively). The pattern persists in the middle oral cavity for these instruments, though attenuated in magnitude and consistency. This pattern, however, is not observed in the posterior oral cavity. In contrast, for the trombone artists, movements in all three areas of the oral cavity are minimal and show no consistent patterns.

Movements of the tongue during the larger 10th interval slur are shown in Figure 5. The trumpet and horn players show the same movement patterns in the anterior oral cavity as in the octave slur, but now with greater magnitude. (+6.38 mm and -5.84 mm/+5.36 and -6.88 for ascending and descending slurs, trumpet/horn, respectively). For this larger interval, only the trumpet players sustain the pattern observed for octave slurs within the middle oral cavity. Again, the trombone players demonstrate very small movements, and no consistent patterns during the exercise. As

with the octave slurs, no consistent patterns are apparent in the posterior oral cavity for any instrument.

Coronal Image Analyses

Average cross-sectional changes in the area of the SLAC during octave and tenth slurs are shown in Figure 6. Columns extending below the center line represent decreases in channel area while those above the center line correspond to increases in channel area. During the ascending phase of octave slurs, a progressive decrease in area is observed in trombone, trumpet and horn players (-131 , -216.6 , and -374.3 mm², smallest to largest change, respectively). During the descending phase, similar directional changes are seen, though of lesser magnitude ($+3.6$, $+197.5$, and $+280.2$ mm² for trombone, trumpet and horn). The amount of change in area for the trumpet and horn players is greatest during the larger interval exercise (trumpet: $-216.6/+197.5$ vs $-225.4/+240$, and horn: $-374.3/+280.2$ vs $-408.6/+346.3$ for octave and tenth intervals,

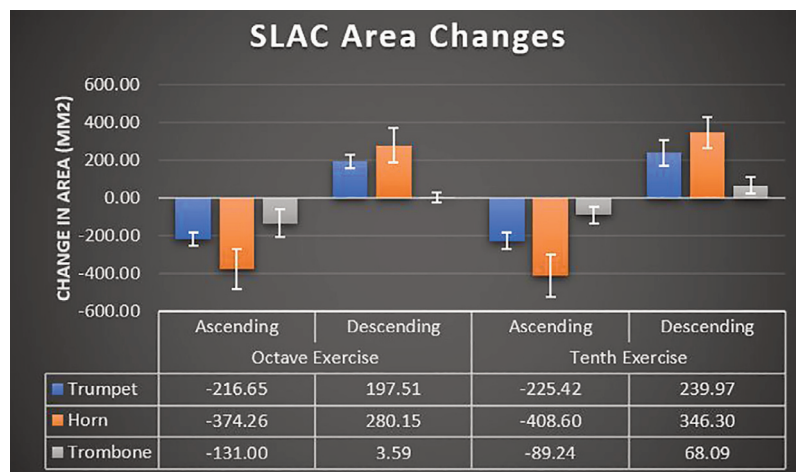


FIGURE 6. Changes in supra-lingual air channel (SLAC) area (mean $\pm$ SE) for each instrument group during the ascending and descending phases of the octave and tenth interval exercises.

respectively). For the trombonists, this interval dependent discrepancy is not replicated.

DISCUSSION

General Observations

Former work by our research group suggests that the performance of large interval slurs in French horn players involves some distinct and replicable patterns in terms of oral cavity motor activity involving the tongue.⁽¹⁴⁾ Specifically, with respect primarily to the anterior region of the oral cavity, there is an upward and forward movement of the tongue during ascending slurs that becomes more pronounced with larger intervals. Further, the height of the SLAC formed between the dorsal aspect of the tongue and the hard palate is commensurately reduced, suggesting a reduction in SLAC area. These patterns are reversed upon playing the descending portion of the interval. The current study re-asserts the existence of these phenomena in horn players, but now provides additional evidence that similar movements occur in trumpet players, but not in trombone players. In addition, this research improves the assessment of changes in the SLAC from our previous study by measuring the cross-sectional area of the channel rather than simply assessing changes in vertical height at the mid-point.⁽¹⁴⁾

Physical Principles Related to Changes in Pitch

Studies examining the physics of brass playing have recently been compiled in a book by Campbell et al. entitled *The Science of Brass Instruments*.⁽²¹⁾ Drawing from this and other publications, several observations can be made. First, the vibration or oscillation frequency of the lips matches the frequency of the pitch being played.⁽²²⁾ Consequently, during large interval slurs, rapid alterations in lip vibration frequency must be accomplished with accuracy and repeatability.

Several factors influence lip oscillation frequency. Among these are changes in lip tension and aperture size between the lips, these being under the influence of multiple facial muscles that regulate embouchure formation.^(15,21,23–25) Additionally, in trumpet players, intra-oral pressure and flow changes can modulate pitch, with higher pressures and flow rates accompanying higher note frequencies.⁽²⁶⁾ For this reason, we have previously contended that reductions in the caliber of the SLAC through movements of the tongue may aid in playing the higher pitches of slurred intervals in horn players.⁽¹⁴⁾ However, other factors may also come into play.

In a study involving trombone players by Fréour and Scavone,⁽²⁷⁾ the authors observe that the oscillations of the lips during note production are sustained by the complex coupling of the flow of air between the lips and resonance characteristics that are created both within the oral cavity,

and in the air-column distal to the lips (i.e., within the mouthpiece/instrument system). They contend that note changes occurring during the performance of interval slurs (ascending arpeggios) may actually be aided by dynamic morphologic modifications of this system, particularly in the very high register (D_5 and F_5). With respect to involvement of the vocal tract, they refer to these conformational changes as vocal tract tuning.

The influence of this proposed vocal tract tuning may have implications for motor activity of the embouchure muscles. The work of Kaburagi et al.⁽²⁸⁾ provides support for this idea. Utilizing a professional trumpet player and a fabricated trumpet constructed of concatenated polyvinyl chloride tubes terminating in a truncated cone, they first had the subject play notes in the low, middle, and high range of the instrument ($E\flat_4$, $B\flat_4$, and F_4 , respectively). (Author's note: $E\flat_4$ is not a naturally occurring harmonic for a conventional $B\flat$ trumpet, but was the lowest sounding frequency obtainable on the device.) Volumetric MRI measurements during the playing of these notes allowed the researchers to quantify conformational changes of the oral cavity. The changes involved a reduction of the vocal tract area in the anterior oral cavity and commensurate increases in the area of the posterior oral cavity as pitch frequency increased. These variables were then used to conduct computer simulation studies employing a lip blowing simulation model developed by Adachi and Sato.⁽²⁹⁾ The simulations indicated that while vibration frequency itself was not directly affected by oral cavity shape, the production of higher pitches could theoretically be accomplished more easily at a lower tension of the lips because of changes in oral cavity shape. Further, oral cavity shape influenced the minimum blowing pressure required to vibrate the lips. They showed that low pitches are more efficiently sounded with the tongue in a low position within the oral cavity and that high notes occur most efficiently when the tongue is raised. Finally, the authors suggest that changes between harmonics are accomplished more efficiently when the oral cavity is modified, requiring less dramatic changes in lung pressure.

Application to the Current Study

In this study, we have demonstrated patterned oral cavity modifications caused by the tongue that are closely tied to changes in pitch during large interval slurs in trumpet and horn players, but not in trombone players. From the brief review of literature cited above, it is reasonable to assume that these conformational changes may have beneficial mechanical influence on pitch modulation that involves, to some extent, what we have referred to as vocal tract tuning. Further, these beneficial effects may allow higher notes in slur sequences to be achieved with lower embouchure muscle tension and lower driving pressure. However, an obvious question relates to our observation that patterned lingual movements were not observed in trombone players during the performance of octave and tenth slurs.



FIGURE 7. *Left*, The harmonic sequence for trumpet in B $\flat$ (first note is the 2nd harmonic, B $\flat_2$ concert). Performed notes were harmonics 2 and 5 for the 10th interval for both trumpet and trombone, though for trombone, the 2nd harmonic is 1 octave lower. *Right*, The harmonic sequence for French horn in E $\flat$ (first note is the fundamental frequency, E $\flat_2$ concert). Performed notes on horn were harmonics 4 and 10 for the 10th interval.

For this study, we restricted the intervals that were compared to octave and tenth intervals commonly encountered in musical literature for these instruments. However, when comparing the relative difficulty of performing these intervals, it is instructive to compare where the performed notes occur in the natural harmonic series of the trumpet, trombone and horn. For the trumpet and trombone players, the largest interval (10th) utilized notes that comprise the 2nd and 5th harmonics, whereas for the horn players, this interval involves the 4th and 10th harmonics. As can be seen in Figure 7, there are 5 naturally occurring notes between these harmonics for the horn, whereas for the trumpet and trombone, there are only 2 between the 2nd and 5th harmonics. Thus, the horn players presumably had a more difficult task in accurately performing these slurs cleanly, hypothetically requiring finer adjustments of the multiple factors affecting vibration frequency of the lips. This may include a more pronounced adjustment of the oral cavity conformation by the tongue, as suggested by the greater SLAC area changes in the horn players depicted in Figure 6 for both intervals. This idea, however, is purely conjecture at this point based upon our empirical observations and should be treated with caution.

As mentioned previously, the exercises performed in the corpus of the MRI Brass Repository have not been identical over the 11 years of its development. For example, in the trombone players examined between 2015–2019 who were used in this study, the large interval slurs that were performed included slurs extending beyond the octave and tenth intervals reported here to include changes up to two octaves (B $\flat_2$ – B $\flat_4$ concert). In terms of the natural harmonic sequence of the trombone, these 2-octave slurs occurred between the 2nd and 8th harmonics. In fact, top trombonists are capable of playing notes up to and including the 12th harmonic. While intervals up to two octaves were within the ability of our trombone players, for trumpet and horn players, they were found to be too difficult on the MRI-compatible instruments, being at the upper extreme of these instrument's range. In reviewing videos of our trombone artists in which these larger intervals were recorded, lingual movement patterns that mimic trumpet and French horn players were clearly seen (see Supplementary Video: <https://youtu.be/akDwq92PnVA?feature=shared>). This finding is consonant with the afore-mentioned work of Fréour and Scavone,⁽²⁷⁾ where vocal tract tuning was most likely occurring in the upper extremes of the trombone range. Our observation that patterned adjustments within the oral

cavity did not occur in trombone players must therefore be interpreted in light of the delimitations imposed by restricting measured intervals to notes that were well below the upper extreme of that instrument's playable range.

Additionally, there are some obvious physical differences between trombones, French horns, and trumpets that may help to explain why horn players and trumpet players employed the movement strategies we have reported while trombonists did not. First among these has to do with mouthpiece bore size. The bore of a mouthpiece refers to the smallest size of the hole running through the middle, usually measured in mm. Trumpet and French horn mouthpieces have similar bore sizes (3.78–4.5 mm and 4.0–4.7 mm, respectively), while trombone mouthpieces are larger (~6.4–7.1 mm). The smaller the size of the bore, the greater is the resistance to air flow through the mouthpiece. The bore diameter of the instruments also affects resistance. Here again, the trumpet and French horn are roughly the same (~11–11.6 mm) while the trombone is ~12.7 mm. It seems feasible to suggest that performing ascending large intervals in smaller bore instruments requires greater effort in overcoming resistance than in larger bore instruments, and that reducing the caliber of the SLAC assists in this process for the smaller bore instruments. Further, returning to the lower note in the interval would show roughly commensurate but opposite changes in SLAC size. It appears that a similar strategy only occurs when larger intervals than the tenth are performed on trombone.

Conclusion and Recommendations

In performing intervals of an octave and a tenth, trumpet and horn players modify the shape of the oral cavity by patterned movements of the tongue that presumably aid in the execution of the task. These modifications have been referred to as vocal tract tuning, and proposed mechanisms for this strategy have been provided by several researchers.^(26–31) In trombonists, these dynamic changes are not seen within the range of the instrument studied in the current investigation, but have been anecdotally demonstrated when the range is extended to the upper limits of the instrument. Future work should be conducted in which a systematic examination of this latter proposal is verified.

Authors' Contributions: Peter Iltis: study design, subject recruitment, data collection and processing, manuscript preparation. Jens Frahm and Dirk Voit: data collection, manuscript review. Aaron Wright and Amy Dever: data processing, manuscript review.

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- 1 anonymous reviewer

Supplementary Video: Professional trombonist playing large intervals up to 2 octaves. Note how tongue movement is minimal until the largest intervals. Available at: <https://youtu.be/akDwg92PnVA?feature=shared>

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